

**Software Development Plan: Project THOR**

Richard Galero

University of Maryland Global Campus

Dr. Michael Pelosi

Nov 2022

## **1. Introduction**

The 48<sup>TH</sup> Maintenance Group based at RAF Lakenheath in the United Kingdom is an aircraft maintenance organization responsible for maintaining the newest fleet of F-35 Lightning II aircraft for the United States Air Forces in Europe. They operate and maintain these aircraft in support of flying operations within the European and African Command.

The F-35 Lightning II is a fifth-generation fighter and attack produced by Lockheed Martin. It is the most lethal, survivable and connected fighter jet in the world (Lockheed Martin, 2022) providing pilots critical advantage and enables them to execute the mission and come home safely. The F-35s have a dedicated information system that enables tracking of maintenance information data for maintenance managers to use in decision-making.

The Autonomic Logistics Information System or ALIS integrates many functional capabilities including operations, maintenance, prognostic, supply chain, customer support services, training, and technical data. However, some of its advanced features face some challenges to export datasets and integrate them to legacy processes. Its inability to export datasets that are required to develop mandated reports lead technicians to result in using manual processes in multiple Excel documents. This challenge is what we aim to solve in this software development project by developing a web application that will be able to serve as collection and reporting tool for maintenance documentation and flight data recording.

## **2. Project details**

To facilitate proper tracking of flight data and maintenance information, a new information system must be implemented to bridge the gap that ALIS was unable to deliver. THOR will enable technicians and managers to directly input data and retrieve or download for analysis. Furthermore, THOR will also allow interfaces to other systems or dashboarding tools such as Microsoft Power BI, Tableau, or Qlik, enabling maintenance managers and analyst to better understand data and leverage it in decision-making processes.

This will cut down manual data processing that is currently done on Excel documents and expedite product delivery by 200%. THOR will cost the organization zero additional cost to the organization because it will be developed organically using already available tools and environment.

## **2.1 Hardware requirements**

- Virtual Machine with Intel® Xeon® Platinum 8171M CPU @ 2.60GHz 210 GHz

## **2.2 Software requirements**

- Python 3 Version 3.11.0 Oct 24, 2022
- Flask Version 2.2.2
- HTML5
- Microsoft Visual Studio Code Version 1.72 September 2022
- Microsoft Visio 2016
- MySQL Version 8.0.30 July 2022
- Microsoft Office 365 App for the Enterprise
- ER Assistant Version 2.10

## **2.3 Scope of project**

Outlined below are specific items that will be delivered by this software development project:

- THOR web application
- MySQL database
- Application program interface to enable connection to Power BI
- Product documentation and manuals
- Entity-Relationship Model

## **2.4 Out of scope**

What will not be within the scope of this project:

- Deployment and maintenance will be hand-off to the US Air Force Helpdesk and will not be within the scope of this project
- Security validation will also be handed to the Air Force Cybersecurity office for testing and validation
- Future upgrades

## **3. Software development process**

This section of the project plan provides a description of the software development life cycle methodology that will be utilized in the implementation of this project. It provides general overview of the method and outlines the different phases. Each event will be discussed in detail and an overview of typical schedule of events. For this software development project, we will be utilizing the SCRUM methodology.

### **3.1 Explanation of the project/management model that will be applied**

In software development, Scrum represents a team working together to achieve a certain goal. It is a simple approach to software development utilizing an iterative model revolving around a set of events, roles, and artifacts (Stellman & Greene, 2017). Events in Scrum includes the sprint, sprint planning, daily scrum, sprint review, and the sprint retrospective which we will discuss more in detail later. The different roles in Scrum include the product owner, scrum master, and the development team. Finally, there are specific artifacts that practitioners of Scrum use, the product backlog, sprint backlog, and the increment. The Scrum framework is illustrated in the diagram below (Figure 1).

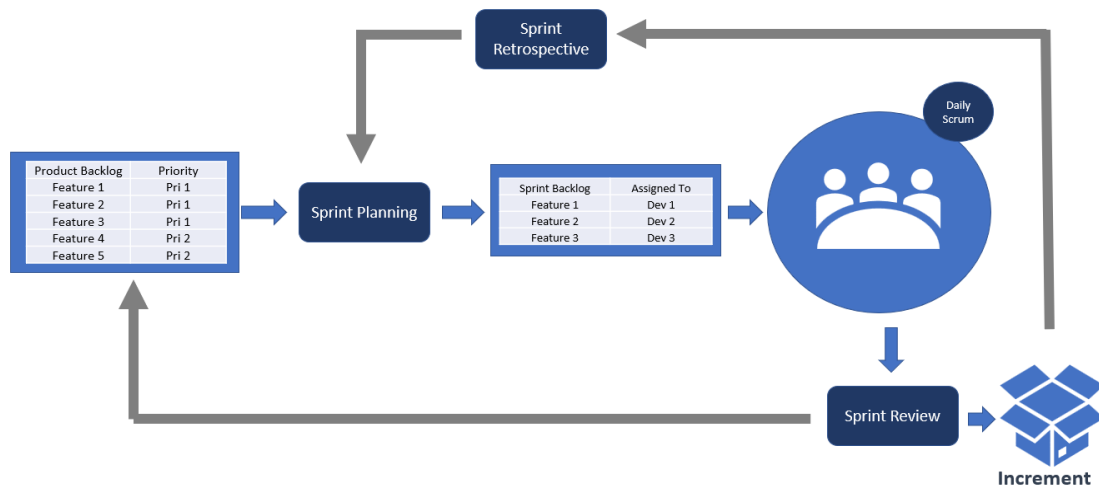


Figure 1. Scrum Framework

### 3.2 Sprint

Sprint is a series of timeboxed event or cycles of development that follows Scrum. During Sprint, our focus is on the Sprint Backlog and assigned tasks that will be realized during the planning meeting. Each person will be working on one item in the Sprint Backlog at a time and his/her focus in on that task. He or she will move on to the next task once that is complete.

### 3.3 Sprint planning

Sprint planning is the Scrum event dedicated to identifying the work that will be accomplished during the Sprint. This is a collaborative work of the entire Scrum team. The Product Owner ensures that attendees are well prepared and ready to tackle the most important Product Backlog items and how it ties to the Product Goal (Scrum.org, 2022). In general, Sprint Planning addresses these three topics:

- Why is this Sprint valuable?
- What can be Done this Sprint?
- How will the chosen work get done?

### **3.4 Daily Scrum**

The purpose of the Daily Scrum is to discuss and inspect the progress of the Sprint Goal and adapt to the Sprint Backlog as necessary to ensure the goal is met in a timely manner. It is a 15-minute event for the Developers for the Scrum team. The Scrum master is responsible to ensure the Daily Scrum happens and guides the development team in how to accomplish the Daily Scrum. Typically, it is consistently held at the same time of the day and in the same location. It is devised to answer these two questions:

- What did you do yesterday?
- What are the challenges you face to do your task today?

### **3.5 Sprint Review**

Sprint review is the penultimate event in a Sprint cycle. Its purpose is to inspect the result of the Sprint and plan for future adaptations. During the review, the team presents their finished work to key stakeholders and the overall progress towards the Product Goal. Based on the result of the Sprint review, the team discusses what to do next. Furthermore, the Product Backlog may be refined to meet new opportunities.

### **3.6 Sprint Retrospective**

The final step in the Sprint cycle is the Retrospective. The purpose of the Sprint Retrospective is to discuss what worked and what did not work. Also, the team plans ways to improve and increase the quality and their effectiveness for future cycles. In retrospectives, the team targets and tries to answer the following questions:

- What went well in the Sprint?
- What could be improved?
- What will we commit to improve the next Sprint?

#### 4. Scrum for Project THOR

This section discusses how we are going to implement the Scrum methodology in the current project.

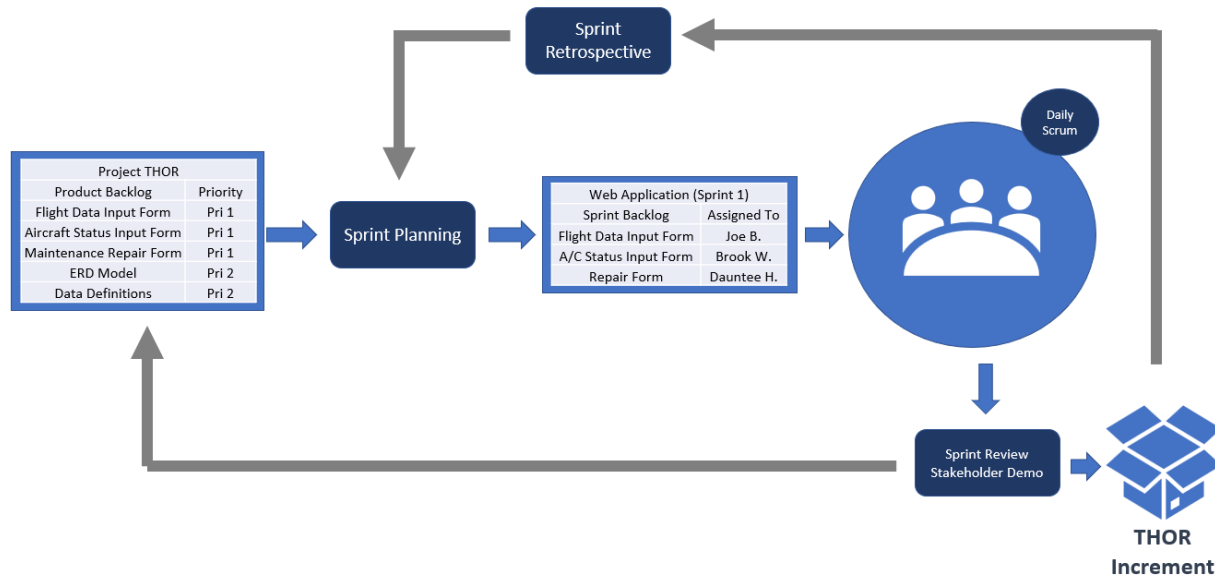


Figure 2. Project THOR Scrum Diagram

##### 4.1 Sprint cycle

The overall sprint planning cycle is a three iteration of 30-day sprints totaling to 90 days of development cycle. In each 30-day sprint, the development team will incrementally build the web application, the database, and the application interfaces needed to enable connectivity and integration of the web app to other systems. The first sprint will be focused on the frontend web application, followed by the creation of the MySQL database and then the API development.

##### 4.2 Daily Scrum and the Development team

The team will be made up of a Product Owner, Scrum Master, and three Developers. The product owner will be responsible for developing the overall Product Goal, creating, prioritizing, and communicating the backlog items, and ensuring the goal is transparent and understood by the entire team. The Scrum Master serves the Scrum team and guides them toward successful implementation of the methodology. The

Scrum master serves as a coach ensuring Scrum best practices are followed. Also, he or she is responsible for the removal of impediments to the Scrum team's success. Lastly, the developers are responsible for the overall development of THOR. The Dev team will consist of full-stack developers that are self-managing and committed to delivering value to the customers.

Daily Scrum meetings will be held following the Scrum guide from Scrum.org. The team will ensure that the focus is on executing best practices of Scrum with the aim to deliver value and deliver often to the stakeholders. The team is responsible to produce a product goal progress tracking mechanism such as Gantt charts to ensure the goal is met in a timely manner. Any issues that may cause delay should be communicated to the Product Owner and reprioritize if needed.

#### **4.3 Sprint review and Stakeholder demonstration**

Upon completion of each Sprint, the team will showcase to the stakeholders what was completed during the Sprint. At a minimum, the Sprint review will be attended by the Scrum Team and key Stakeholders. The Dev team will discuss what was completed and what was not completed along with what went well and the problems they ran into during the Sprint. The overall Product Backlog will also be discussed by the Product Owner and targets a delivery date based on the progress of the project.

Furthermore, the team collaborates with the stakeholders on the feedback and implement changes to the next Sprint planning if necessary. Lastly, the timeline, budget, potential capabilities, will also be reviewed and discussed.

#### **4.4 Planning for issues and subsequent Sprint cycles**

The goal of each 30-day Sprint is to complete each module of the Product, web application, backend database, and application program interfaces. However, problems may arise during the Sprint and must be communicated effectively to the Product Owner and Stakeholders. If there will be anticipated delays due to an issue with the code or other problem with the development process, it is best prioritized what can be done during the Sprint and save the problematic module for later or for the next Sprint.



## 4.5 Sprint backlog

|                        |                     |                   |                 |                         |                            |  |
|------------------------|---------------------|-------------------|-----------------|-------------------------|----------------------------|--|
| <b>PROJECT NAME</b>    | <b>Project THOR</b> | <b>START DATE</b> | <b>END DATE</b> | <b>OVERALL PROGRESS</b> | <b>PROJECT DELIVERABLE</b> |  |
| <b>PRODUCT MANAGER</b> | Richard Galero      | 2-Jan-23          | 14-Apr-23       | 0%                      | <b>SCOPE STATEMENT</b>     |  |

| AT RISK | TASK NAME | FEATURE TYPE | RESPONSIBLE | STORY POINTS | START     | FINISH    | DURATION in days | STATUS      | COMMENTS |
|---------|-----------|--------------|-------------|--------------|-----------|-----------|------------------|-------------|----------|
|         | SPRINT 1  |              |             |              | 2-Jan-23  | 31-Jan-23 | 10               | Not Started |          |
|         | Feature 1 |              | Joe B.      |              |           |           | 4                | Not Started |          |
|         | Feature 2 |              | Brook W.    |              |           |           | 5                | Not Started |          |
|         | Feature 3 |              | Dauntee H.  |              |           |           | 4                | Not Started |          |
|         | SPRINT 2  |              |             |              | 6-Feb-23  | 5-Mar-23  | 8                | Not Started |          |
|         | Feature 4 |              | Joe B.      |              |           |           | 1                | Not Started |          |
|         | Feature 5 |              | Brook W.    |              |           |           | 4                | Not Started |          |
|         | Feature 6 |              | Dauntee H.  |              |           |           | 2                | Not Started |          |
|         | SPRINT 3  |              |             |              | 13-Mar-23 | 14-Apr-23 | 10               | Not Started |          |
|         | Feature 7 |              | Joe B.      |              |           |           | 4                | Not Started |          |
|         | Feature 8 |              | Brook W.    |              |           |           | 5                | Not Started |          |
|         | Feature 9 |              | Dauntee H.  |              |           |           | 3                | Not Started |          |

| STATUS      |
|-------------|
| Complete    |
| Overdue     |
| In Progress |
| Not Started |
| On Hold     |

Figure 3. Sprint Backlog

4.1 Sample Sprint Schedule

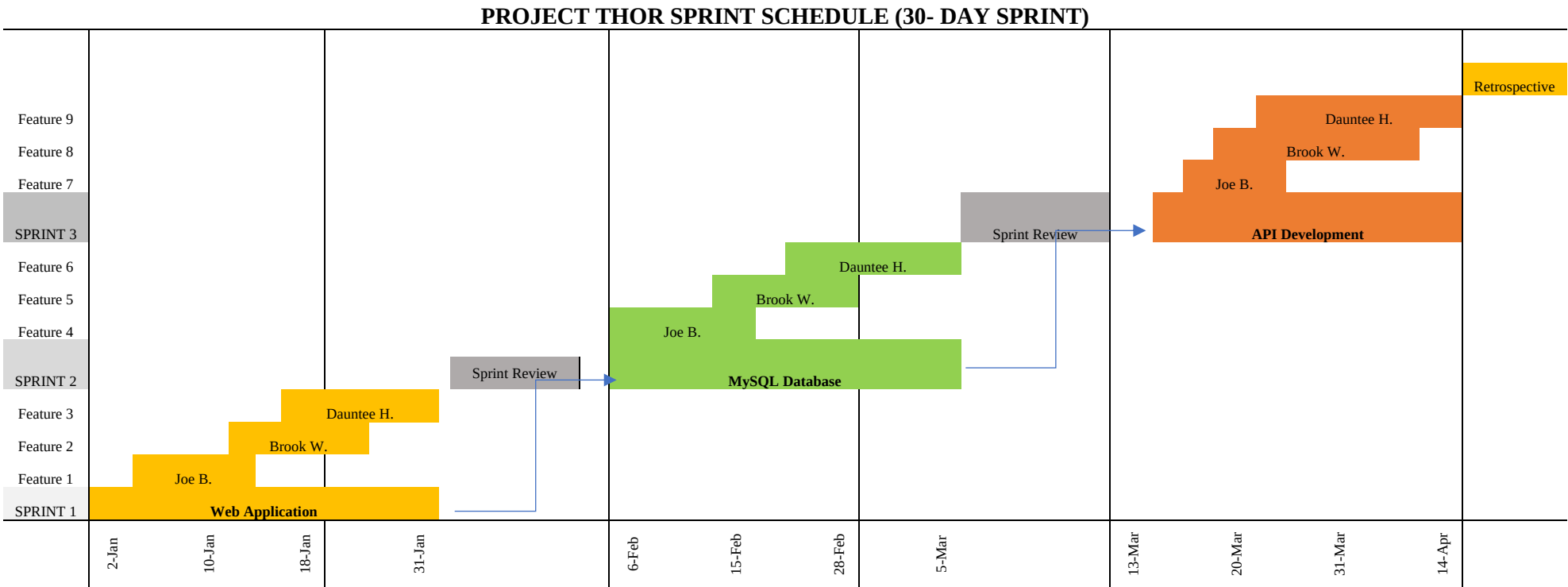


Figure 4. Sprint Schedule (Gannt Chart)

## **5. Risk Analysis**

Like any other project, there will be some risk associated with a software development project. In this section we will briefly discuss some of the risk that we may encounter during this project. We will also provide potential action plan to mitigate the risks that we identify. The Indeed Editorial Team published 12 Risks in Software Development (2021). We think some of these risks apply to the Project THOR as well.

### **5.1 Potential risks**

- Code issues
- Aggressive deadlines
- Unmet expectations
- Low productivity
- Stakeholder issues
- Scope creep

### **5.2 Risk mitigation plan**

- Testing code frequently
- Implement coding standards and using best practices
- Clearly communicate with the Product Owner and Stakeholders to ensure deadlines are achievable
- Set good goals using SMART technique to mitigate unmet expectations and low productivity
- Communicate with stakeholders frequently
- Set clear project goals
- Scrum practices are designed to avoid scope creep – follow best practices and Scrum guidelines

## 6. Conclusion

This software development plan aims to give a high-level plan in developing the THOR web application. THOR is a web application that will be used by aircraft maintenance technician in recording flight data information and maintenance status as well as keep historical data for future analyses. THOR has three components, the front-end web application, a MySQL backend database, and application program interface (API) that will enable THOR to feed and make connections to analysis reports and dashboards build in Microsoft Power BI program. THOR will enable a speedy download of historical data that the ALIS system currently lacks for managers and analysts to use in decisions. It will be developed and implemented using the Scrum methodology which is an agile and iterative approach to software development.

Scrum embodies cross-functional team working towards a common goal of developing and delivering working software to fir the needs of the customer. The decision to use Scrum for this project is a no-brainer as it perfectly aligns with the current Department of Defense standard of moving away from traditional and restrictive methodologies such as Waterfall.

Compared to Waterfall, Scrum is less restrictive and enables developers and stakeholders to collaborate and implement feedback to ensure the product aligns with the goal. Fostering communication between teams, Scrum will ensure that the scope of the project stay focus and does not cause additional time and resources. And with the scope of the THOR project, we anticipate some challenges and risks during the development and designed a plan to mitigate these risks. These kinds of complicated and complex problems are where Scrum is best suited according to the PMI Agile Practice Guide (2021). Scrum provides the ability to break down complex projects into small and much more manageable pieces and that is why it is a great fit for the THOR project. Furthermore, Scrum is also better suited for small teams such as the Project THOR development team which consists of five to 8 members.

Scrum enables self-governing teams and empower them to make the decision best suited for the project.

THOR's small cross-functional team is better suited to operate in a Scrum environment compared to other software development methodology that requires a much bigger team.

Finally, Scrum's short iteration cycles will enable that our stakeholders are well abreast of the status of the project. By leveraging feedback through consistent communication between the development team and stakeholders, we will be able to ensure that the project can adjust to evolving requirements to meet the customers' needs. For these many positive reasons that we selected Scrum methodology to build and implement Project THOR.

## References

Denning, S. (2015). Agile: The World's Most Popular Innovation Engine. Retrieved from <https://www.forbes.com/sites/stevedenning/2015/07/23/the-worlds-most-popular-innovation-engine/>

Indeed, (2021). *12 Risks in Software Development*. Retrieved from <https://www.indeed.com/career-advice/career-development/risks-in-software-development>

Lockheed Martin (2022). F-35 Lightning II. Retrieved from <https://www.lockheedmartin.com/en-us/products/f-35.html>

Scrum.org, (2022). What is Scrum? A better way to work together and get work done. Retrieved from <https://www.scrum.org/resources/what-is-scrum>

Stellman, A., & Greene, J. (2017). *Head First Agile*. O'reilly Media.

Value Insights. (2021). When NOT to use Scrum. Retrieved from <https://valueinsights.ch/-when-not-to-use-scrum/>